

Executive Summary

Forests dominate Finland's landscape, covering approximately 75 percent of the land area, or about 23 million hectares. This makes Finland the most forested country in Europe by proportion of land area. These forests form the foundation of the Finnish bioeconomy, supporting major industries in timber, pulp and paper, and bioenergy production. However, Finland's forests have recently reached an ecological inflection point. We were tasked to design a sovereign wealth fund that channels forest linked revenues into assets that help biodiversity to prosper, while meeting long-term returns for the fund.

We have looked at key investment philosophies that will help the BW fund to maintain its dual mandate that recognizes stability in financial markets and also taking into consideration national prosperity key to its dual mandate. We believe there are three key philosophies guarding the fund the first is sustainable wealth creation aimed at creating long-term growth for the fund. The second is transparency and measurability that are key to understand which KPIs to understand and look at together a better understanding of the biodiversity impact for the fund. We also considered diversification and stewardship as key to the fund to help maintain exposure to multiple asset classes resulting in a balanced return.

Focusing on sustainable screening by creating the first sustainability-based filter that includes sector and activities exclusion baseline ESG compliance and preliminary biodiversity impact screening and evaluate those screen companies based on financial performance biodiversity assessment and nature-related risk assessment leading to asset allocation focused on ESG.

We looked at the current allocation and redesigned the allocation strategically into real assets, public and private credit, fixed income, conservation and restoration assets and cash for providing liquidity and returns in the long run while focusing on biodiversity conservation by transitioning from the current allocation into the new allocation in 3 phases from 2025 – 2035. We will focus on establishing the foundation for nature-aligned investments while maintaining liquidity and portfolio stability with real assets and fixed income allocation increasing by 300 bps and equities being reduced by 600 bps while taking into consideration a strong biodiversity scoring framework, with every new investment evaluated against minimum nature-impact criteria.

2. Problems/Background

2.1. Finland's Natural Capital and Ecological Context

Forests dominate Finland's landscape, covering approximately 75 percent of the land area, or about 23 million hectares. This makes Finland the most forested country in Europe by proportion of land area. These forests form the foundation of the Finnish bioeconomy, supporting major industries in timber, pulp and paper, and bioenergy production.

However, Finland's forests have recently reached an ecological inflection point. According to Statistics Finland (2023) and the Natural Resources Institute Finland (Luke), Finnish forests became a net carbon source in 2021 for the first time in recorded history. This reversal was driven by higher logging volumes, declining forest growth rates, and increasing soil emissions linked to rising temperatures.

A 2025 WWF Finland report estimates that only 10 percent of Finland's forests are strictly protected, with just 3 percent in the southern regions, which contain the highest biodiversity density. Much of the remaining old-growth forest is fragmented and lies outside protected zones, making it susceptible to industrial harvesting.

2.2. Economic Dependence and Structural Risks

Forestry and bio-based industries remain a cornerstone of Finland's economy, accounting for 19% of industrial output and around 20% of export revenues according to the Finnish Forest Industries Federation (2024). However, this reliance on forest-based production introduces vulnerabilities to both global commodity cycles and climate-related regulations.

In 2021, Finland lost its carbon sink status for the first time. The land use, land-use change, and forestry (LULUCF) sector recorded net emissions rather than net removals, marking a structural shift driven by intensive harvesting, aging forest stands, and increasing soil emissions. This development highlights a fundamental imbalance between the extraction rate and forest regeneration.

If current harvesting and soil carbon loss trends continue, Finland's forests may remain a **net carbon source** throughout the 2030s, limiting the country's ability to meet its EU climate targets and increasing exposure to carbon compliance costs under the **EU Emissions Trading System (ETS)**.

This shift also signals growing financial risks for the forestry sector. Biodiversity degradation can impair asset values, limit future harvest potential, and reduce the creditworthiness of the forest-linked investments. As a result, the long-term competitiveness of Finland's bioeconomy increasingly depends on transitioning towards sustainable land use and biodiversity-positive growth.

2.3. Finland's Biodiversity Commitments

Finland is globally recognized for its leadership in climate policy and emissions reduction. In 2019, it became the first country to legislate a 2035 net-zero target, embedded in the Climate Change Act of 2022, which introduced a binding carbon budgeting system and independent scientific oversight. Programs like METSO were introduced that help Finland's landowners reduce the loss of biodiversity on their lands and provide permanent protection for the conservation of land.

Emissions and removals in the LULUCF sector by land use category (Mt CO₂ eq)
Positive figures mean emissions and negative figures mean removals (sink)

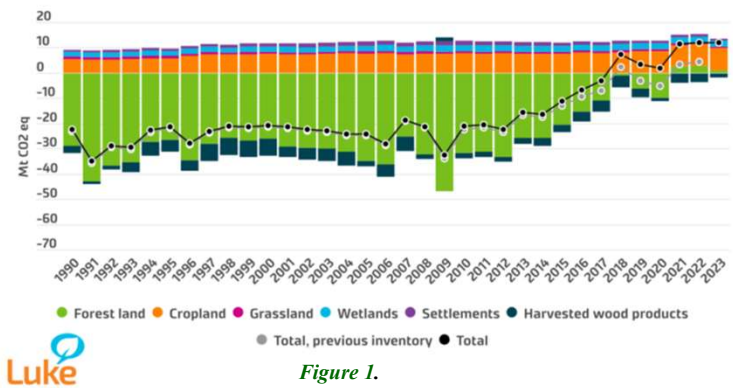


Figure 1.

While programs have helped Finland reduce some of its declining biodiversity, stakeholders caution that restoring Finland's full biodiversity loss will require an order-of-magnitude more capital to catalyze co-investment, develop

2.4. The Role of the Borealis Wealth Fund

2.4.1 Strategic Purpose

The Borealis Wealth Fund was launched in 2021 as Finland's flagship sovereign wealth vehicle, created to align ecological goals with economic sustainability. The fund's core objective is to transform Finland's forestry-linked revenues into biodiversity-enhancing assets while maintaining stable long-term financial returns.

Initially, BWF pursued a broad ESG agenda spanning green infrastructure, low-carbon technologies, and international mandates, but these early efforts failed to directly address Finland's most pressing ecological challenge, the collapse of national identity.

2.4.2 Mandate Realignment

In 2025, following EU biodiversity regulations and mounting ecological data, the Finnish government refocused BWF's strategy toward biodiversity preservation and natural capital regeneration. This pivot includes:

- Forest ecosystem restoration
- Peatland and wetland recovery
- Habitat protection and species conservation

This realignment is not only environmental but also economically strategic, as the degradation of forest carbon sinks threatens Finland's forestry sector, export credibility, and asset valuations. Thus, BWF comes in as a stabilizer, redirecting forestry cash flows toward sustainable capital formation and biodiversity restoration efforts.

The BWF is capitalized through a 5 percent allocation from gross forestry sector revenues, approximately €1 billion per year, and it currently manages €10 billion across equity and fixed income. The investment strategy must balance liquidity, risk, and biodiversity-aligned opportunities, and must convert forestry cash flows into nature-positive capital formation at scale.

2.4.3 Capitalization and Mandate

BWF sovereign fund is unique when it comes to funding, as most of the sovereign wealth funds (SWFs) are funded using oil or surplus reserves, whereas the BWF is funded by a 5% levy on gross forestry sector revenues (~€1B/year during the year of 2025), totaling around €10B AUM across equities and fixed income. This creates a unique dual mandate for the BWF:

1. Safeguard Finland's ecological future by reversing biodiversity decline.
2. Ensure financial stability and positive long-term returns from forestry-derived capital.

3. Investment Philosophy and Framework

3.1. Investment Philosophy

The Borealis Wealth Fund (BWF) follows a dual mandate which is to deliver long-term, risk-adjusted financial returns while positively impacting biodiversity and climate resilience in Finland and beyond. This philosophy recognizes that the stability of financial markets and national prosperity are inseparable from the health of natural ecosystem.

The Fund's investment strategy is built on the principle that nature is a productive asset class. Biodiversity, soil fertility, clean water and carbon sequestration represent real economic value that can be measured, monetized and enhanced through disciplined investment. In the BWF's case, biodiversity preservation is not a philanthropic goal but a means of sustaining long-term asset productivity and preserving national identity.

BWF's philosophy is grounded in three guiding beliefs:

1. Sustainable Value Creation

The Fund views sustainability as a driver of growth rather than a constraint. Investments in ecosystem restoration, regenerative agriculture and low carbon infrastructure is deemed to create long-term sustainable growth and mitigate systematic risks to national wealth.

2. Transparency and Measurability

Every investment must demonstrate a quantifiable biodiversity and climate outcome, tracked through standardized indicators. The Fund commits to reporting in alignment with the Taskforce on Nature-related Financial Disclosures (TNFD) and the EU Sustainable Finance Disclosure Regulation (SFDR) to ensure credibility and comparability across asset classes.

3. Diversification and Stewardship

The Fund maintains a balanced exposure across real assets, fixed income, and equities to ensure liquidity and capital preservation. Stewardship is exercised through active ownership, policy engagement, and collaboration with industry partners to embed biodiversity standards in financial markets.

BWF operates with the conviction that long-term returns are maximized while providing a positive biodiversity impact across the globe. By embedding bio-diversity metrics into the core investment philosophy, the Fund transforms biodiversity-positive actions into a competitive edge.

3.2. Biodiversity Impact Measurement Framework

The Borealis Wealth Fund (BWF) measures its environmental performance through a structured Biodiversity Impact Framework designed to quantify ecological outcomes across sectors and asset classes. This framework provides a consistent method for evaluating investments, ensuring that biodiversity and climate performance are measurable, comparable, and verifiable.

Five core indicators are used to evaluate each investment. Together, they produce a Biodiversity Impact Score (BIS) that captures both ecological and financial alignment with the Fund's mandate.

3.2.1 Biodiversity Outcomes

This indicator measures how much of an investment directly enhances or protects biodiversity. It captures land or habitat under protection, restoration projects, or ecosystem services supported by the asset. In forestry, it may reflect the percentage of managed land under active restoration. In infrastructure, it could represent habitat corridors or natural flood mitigation integrated into project design. It quantifies the Fund's most direct contribution to ecological recovery and preservation.

3.2.2 Climate Co-benefits

This indicator evaluates whether an asset functions as a carbon sink or source, capturing its total contribution to climate mitigation. It integrates carbon sequestration and emissions, ensuring that biodiversity and climate goals reinforce one another.

3.2.3 Sustainable Management Practices

This indicator measures how much of the Fund's managed area or portfolio adheres to credible sustainability certifications and operational standards. Examples include FSC or PEFC certifications for forestry, regenerative agriculture verification in agri-tech, or ISO 14001 standards for infrastructure assets.

3.2.4 Governance and Disclosure

This indicator captures how transparent and accountable each investment is in managing biodiversity-related risks. It evaluates the presence of biodiversity policies, alignment with TNFD or SFDR frameworks, independent reporting assurance, and stakeholder engagement.

3.2.5 Nature Related Risk

This indicator assesses an asset's dependence on or vulnerability to nature, such as exposure to deforestation, soil degradation, or regulatory tightening. It contextualizes risk by comparing each investment's exposure to sectoral or geographic benchmarks.

3.2.6 Biodiversity Impact Score

Each indicator is weighted according to its strategic importance, producing a composite Biodiversity Impact Score (BIS). The BIS functions as a unified metric of ecological and financial alignment.

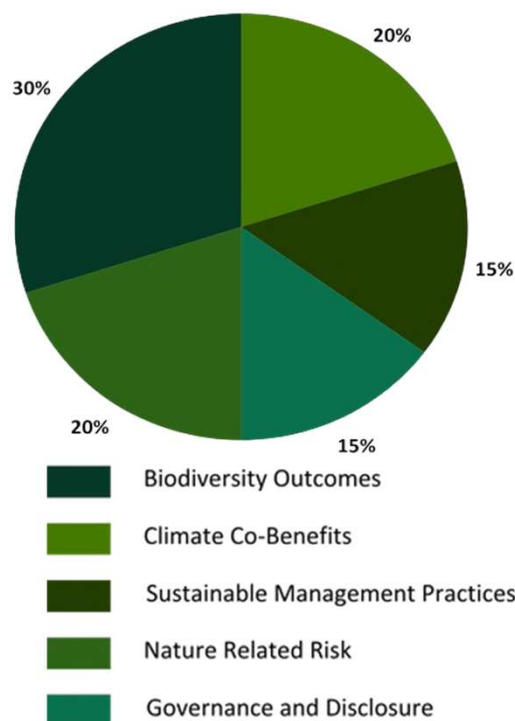


Figure 2.

The Biodiversity Impact Score provides a single measure of an asset's eligibility for investment.

- Assets with scores above 60 are considered investable, indicating strong alignment with biodiversity and climate objectives.
- Assets scoring below 60 are excluded unless a credible improvement plan is in place.

This approach ensures that the Fund's investment decisions are guided by clear, measurable, and science-based thresholds. It embeds biodiversity as a core component of financial due diligence and portfolio construction.

3.3. ESG Integration in the Investment Process

The Borealis Wealth Fund (BWF) embeds environmental, social, and governance (ESG) principles into every stage of its investment process. The goal is to ensure that all capital deployed contributes both to long-term financial performance and to measurable biodiversity and climate outcomes. The integration process is structured into four stages: screening, evaluation, allocation, and monitoring and stewardship. Each stage applies the Biodiversity Impact Score (BIS) as a central decision tool to ensure alignment with the Fund's dual mandate.

3.3.1 Screening

The screening stage serves as the Fund's first filter for potential investments based on their ecological and governance integrity. This process ensures that only assets with a credible potential to generate positive biodiversity outcomes are advanced to further evaluations.

The screening process includes:

1. Sector and Activity Exclusion

The Fund excludes investments that cause significant harm to biodiversity or contribute to ecosystem degradation. Activities associated with deforestation, unsustainable mining, industrial-scale monocropping, or large-scale land-use conversion are automatically screened out. Additionally, projects located within ecologically sensitive zones without mitigation plans are deemed ineligible.

2. Baseline ESG Compliance

Prospective investments must meet a minimum ESG compliance standard aligned with the EU Taxonomy and the Sustainable Finance Disclosure Regulation (SFDR). This includes demonstrating environmental risk management systems, legal compliance, and governance structures that enable responsible oversight.

3. Preliminary Biodiversity Impact Screening

Each investment undergoes a rapid assessment of its likely biodiversity outcomes using sectoral benchmarks and available disclosures. This generates a preliminary Biodiversity Impact Score (BIS), identifying whether the opportunity could realistically exceed the Fund's minimum investability threshold. Only investments that exceeds a preliminary BIS above 60 out of 100 progress to the next phase of evaluations.

Only investments that meet all the 3 screening criteria are to move onto the evaluation process.

3.3.2 Evaluation

Once screened, qualifying investments undergo a comprehensive evaluation that integrates financial, ecological, and governance analyses. This stage balances traditional investment rigor with science-based impact assessment.

The evaluation process includes:

• Financial Performance Assessment

Conventional valuation methodologies such as Discounted Cash Flow (DCF), Internal Rate of Return (IRR), and risk-adjusted return analysis are applied to determine the investment's expected profitability, liquidity, and volatility characteristics. The Fund's financial team ensures that target return thresholds are consistent with long-term fund objectives.

• Biodiversity and Climate Assessment

The Fund conducts a detailed analysis of the five biodiversity indicators defined in Section 3.2, measuring biodiversity outcomes, climate co-benefits, management practices, governance, and nature-related risks. Each indicator is scored quantitatively using verifiable data sources, satellite monitoring, and third-party certifications when available. The outputs of these analyses are combined to produce a refined BIS, which quantifies the asset's ecological contribution alongside financial returns.

• Nature-Related Risk Evaluation

The Fund employs a TNFD-aligned risk framework to identify physical, transition, and systemic risks associated with biodiversity loss. This includes evaluating exposure to climate change impacts, regulatory tightening, and supply chain dependencies on natural capital.

Only investments that demonstrate both a financially viable return profile and a refined BIS of at least 60 are eligible for inclusion in the portfolio. Those exceeding 75 are prioritized as strategic opportunities with strong biodiversity alignment.

3.3.3 Allocation

Allocation decisions translate the evaluation results into capital deployment. The objective is to maintain portfolio balance across asset classes, ensuring diversification, liquidity, and risk-adjusted returns while maximizing biodiversity impact.

The allocation process considers:

• Target Asset Class Weights

Each investment is categorized according to the Fund's strategic allocation defined in Section 4. Real assets, particularly forestry, land holdings, and green infrastructure, serve as the core of the portfolio due to their direct biodiversity and carbon benefits. Complementary exposures in green bonds, private credit, and equities ensure liquidity and steady income flows.

• Performance/Impact Optimization

Capital is preferentially allocated to investments with high BIS scores and low biodiversity-related risks. This ensures that ecological performance enhances rather than dilutes portfolio stability. Projects that score near the investability threshold may be structured with sustainability-linked terms, for example, performance-based coupons or milestone-based equity tranches tied to biodiversity outcomes.

• Dynamic Reallocation

The Fund maintains flexibility to adjust capital distribution as market conditions and ecological data evolve. Reinvestments and new allocations are guided by updated BIS assessments, sectoral biodiversity metrics, and annual performance reviews.

3.3.4 Monitoring and Stewardship

The monitoring phase ensures that biodiversity and financial objectives remain aligned over the investment lifecycle. Continuous oversight reinforces accountability, enables adaptive management, and maintains credibility with stakeholders.

The Fund's monitoring and stewardship framework includes:

- **Annual Biodiversity and ESG Reporting**
All investee companies and projects are required to submit annual disclosures covering biodiversity outcomes, carbon performance, and governance practices. Reporting aligns with TNFD, SFDR, and the Global Reporting Initiative (GRI) biodiversity metrics.
- **Active Performance Tracking**
The internal sustainability team recalculates BIS scores annually based on updated data. Assets showing improvement in biodiversity outcomes may be rewarded with increased capital or performance incentives, while underperforming assets are flagged for review or divestment.
- **Engagement and Capacity Building**
BWF engages directly with portfolio companies, local communities, and policymakers to strengthen biodiversity management practices. Engagement may include technical support, capacity building, or collaboration on restoration and conservation projects.
- **Adaptive Management and Exit Strategy**
Assets failing to meet biodiversity performance thresholds after repeated engagement are subject to structured divestment. This disciplined approach preserves the Fund's integrity and ensures that all portfolio activities contribute positively to the national biodiversity agenda.

4. Asset Allocation

4.1 Current Allocation

The current allocation for 10 billion portfolio is strategically diversified among different asset classes to balance long term goal of sustainability and having returns. The fund has the largest allocation of 4 billion is in global equities screened for ESG criteria, ensuring exposure to sustainable public companies. Real assets make up of 3.5 billion giving exposure to green infrastructure and timberland investments in Finland and international assets. To maintain flexibility and meet liquidity needs, 1 billion is held in cash. Geographically, the portfolio is weighted toward Finland (55%) and the broader European Union (30%), with additional exposure to North America (10%) and global emerging markets (5%).

4.2 New Asset Allocation

4.2.1 Real Assets

Due to the dual mandate, we believe investing in real assets is the most efficient way for capital allocation which provides enough fiscal support to the preservation efforts. This would include renewable energy, sustainable transportation, water infrastructure. This bucket helps the fund diversify away from the forestry sector and focus more on the adjacent industries which require the same attention. This would be the core anchor of our biodiversity strategy, as this provide direct

control over forests, peatlands and wetlands which require special attention especially during this time. This also helps Finland regain its footing as a carbon sink nation.

Target Weights Vs Current Weights

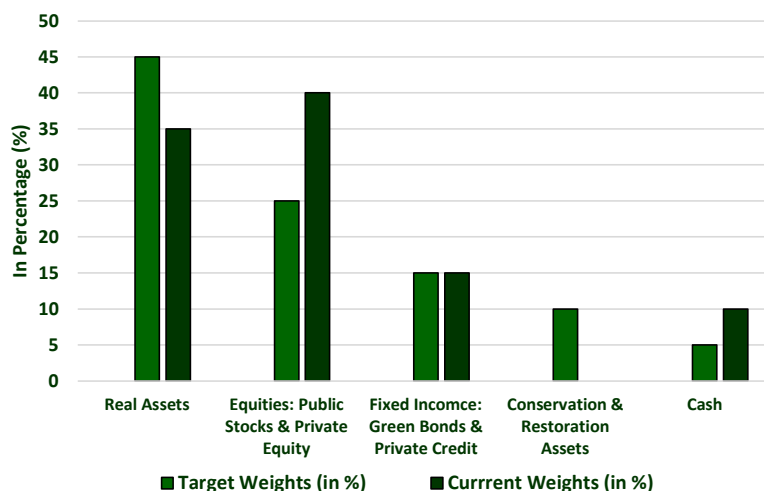


Figure 3.

4.2.2 Public and Private Equities

A global involvement in sustainable economies coupled with innovation, impact and technology. A portion of the fund shall be allocated to private equities to influence modernization and create urgency. The fund believes in circular economies and agriculture-tech explorations to help discoveries and implementation of renewables and efficient soil and waste management techniques. A stake in public equities provides us with liquidity and a growth structure with global leaders in ESG and sustainable management. We aim to divert concentration from Finland to diversify our portfolio by supporting needful geographical areas all while maintaining growth, transparency and a position in global development initiatives.

4.2.3 Fixed Income

Investing in green bonds supports environmentally friendly projects. This can lead to significant positive changes, such as reduced carbon emissions, conservation of natural resources, and the promotion of renewable energy sources. The proceeds from green bond issues are allocated exclusively to environmental projects and initiatives promoting greener practices. In recent years, these bonds have gained significant momentum, becoming an effective tool for investors to achieve environmental impact, diversification, and financial returns. We also want to invest in Private credit as this provides a catalyst to mobilize large pools of Capital and will let us write enforceable biodiversity Covenants to more sustainable returns.

4.2.4 Conservation and Restoration Assets

Allocations in conservation and restoration projects directly focuses on increased biodiversity, halting further degradation, restoration of species and enhancing carbon sequestration practices giving the land resilience and stability. Our aim to is to invest in sustainable forestry management including soil preparation, systematic water frameworks and efficient timber harvesting. Effective investing in this asset class can help Finland bridge the gap between the wildlife they long for while also stabilizing natural conditions for better yields, conservation credits, cost-neutral harvesting and attractive ecological tourism. Involvement in this asset class justifies our contributions proving legitimacy and making the fund centered towards biodiversity.

4.2.5 Cash

Real and solid liquidity to provide a buffer and stability to maintain flexibility in operations, tackle risks and grab new opportunities.

4.3. Transition Plan

The Borealis Wealth Fund (BWF) will implement its transition toward the target allocation through a phased, data-driven approach that aligns annual capital inflows with both financial performance and biodiversity goals. The plan assumes a steady annual capital inflow of approximately €1 billion, with compounded returns resulting in total assets under management (AUM) reaching between €35–40 billion by 2035, depending on realized investment performance.

The transition is designed to achieve a balance between liquidity, stability, and ecological impact, while minimizing disruption to existing positions. Each asset class expands or contracts in a sequenced manner based on market maturity, biodiversity contribution, and the Fund's liquidity requirements.

Under the base-case growth scenario, assets under management are projected to increase from €11.6 billion in 2025 to €37.8 billion by 2035, delivering an estimated CAGR of 6.8–7.2 percent.

4.3.1. Phase 1: Foundation & Strategic Alignment (2025–2027)

The initial phase will focus on establishing the foundation for nature-aligned investments while maintaining liquidity and portfolio stability. Real assets will grow from 35 percent to 38 percent of total holdings, with the majority directed toward green infrastructure and early-stage forestry investments. Green infrastructure will hold steady at 20 percent, targeting renewable energy, water networks, and low-carbon transport projects. Forestry and land holdings will expand from 15 percent to 18 percent as the fund begins acquiring certified sustainable forests and restoration sites in Finland and the European Union.

Fixed income will increase slightly from 15 to 18 percent, as the fund introduces private credit lending to biodiversity-linked enterprises. Public equities will be gradually reduced from 40 to 34 percent, while private equity allocations reach 7 percent by 2027, focusing on ventures in agritech, circular materials, and bioeconomy innovation. Conservation and restoration assets will be introduced at 2 percent by 2028, marking the start of natural capital investments that directly contribute to Finland's ecological recovery.

By 2027, the portfolio is expected to reach approximately €15 billion in assets under management. At this stage, the fund will have established its biodiversity scoring framework, with every new investment evaluated against minimum nature-impact criteria.

4.3.2. Phase II: Scaling and Diversification (2028–2031)

As the foundation is established, the next phase will emphasize scaling investments in forestry, private credit, and restoration assets. The share of real assets will rise from 39 to 44 percent, driven by the expansion of forestry and land holdings to 24 percent. Green infrastructure will remain stable at 20 percent but will transition toward more mature, income-generating renewable portfolios.

Fixed income exposure will stay around 18 percent, though green bonds will decline as private credit expands to nearly 10 percent, funding climate-resilient and biodiversity-positive projects. Equity exposure will moderate to 30 percent, reflecting the continued shift from public to private markets. Public equities will fall to 20 percent, while private equity grows to 10–12 percent, targeting emerging impact ventures and technology enablers for sustainable industries.

Phase 1 (2025–2027)

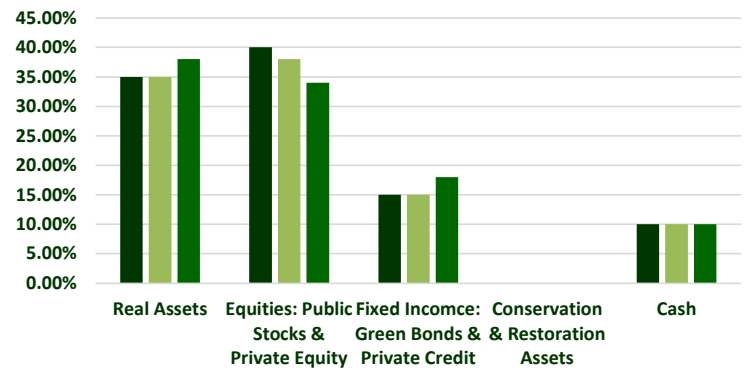


Figure 4. ■ 2025 Weights ■ 2026 Weights ■ 2027 Weights

Conservation and restoration assets will reach 5 percent, expanding to include biodiversity credit markets, peatland rehabilitation, and nature real estate opportunities. Cash reserves will fall to 3–5 percent as the portfolio achieves greater diversification. By 2031, assets under management are projected to exceed €24 billion, with more than half of the portfolio fully aligned with the biodiversity mandate.

Phase 2 (2028–2031)

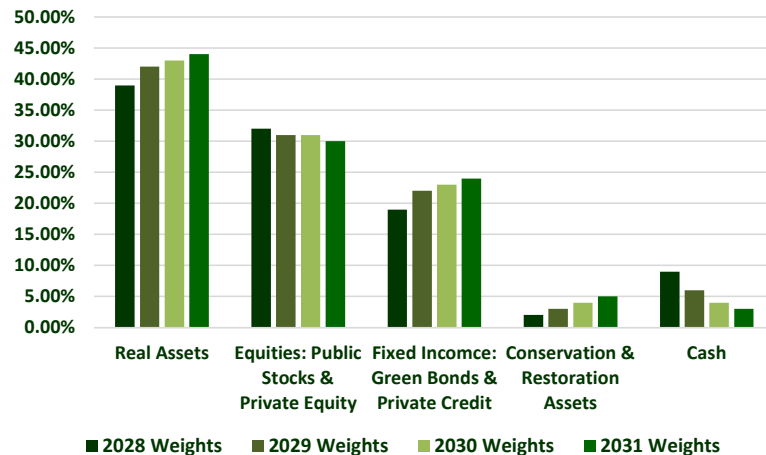


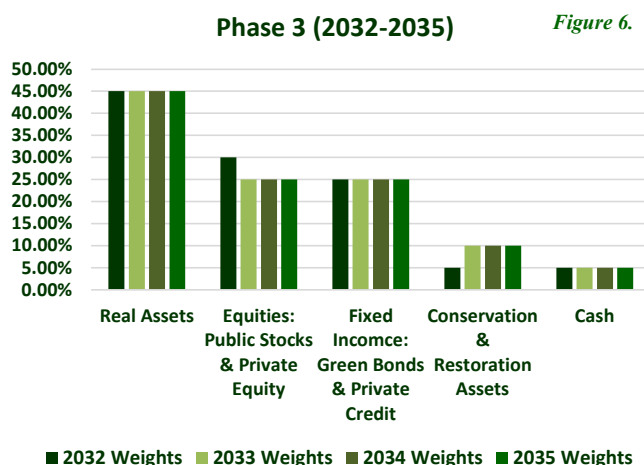
Figure 5.

4.3.3. Phase III: Consolidation and Full Alignment (2032–2035)

The final phase will complete the transition toward the target structure. Real assets will stabilize at 45 percent of the portfolio, led by 25 percent in forestry and land holdings and 20 percent in green infrastructure. Fixed income will remain at 15 percent, with private credit providing the bulk of steady income generation.

Equities will be balanced at 25 percent, with 15 percent in private equity and 10 percent in public equities. Private investments will focus on circular economy solutions, bioproducts, and sustainable manufacturing. Conservation and restoration assets will reach the full 10 percent allocation, representing Finland's long-term investment in natural capital and biodiversity credit markets. Cash will stabilize at 5 percent to maintain flexibility and meet liquidity needs.

By 2035, total assets under management are expected to reach approximately €37–38 billion. The portfolio will be fully aligned with its biodiversity and financial objectives, with an average Biodiversity Impact Score above 70 and an estimated annual return between 7 and 7.5 percent.



The transition will be managed by the Transition Oversight Committee, which will report directly to the Borealis Wealth Fund Board of Trustees. The Committee will oversee all rebalancing decisions, approve capital allocations, and monitor the implementation of biodiversity metrics. Annual transition reports will be published to maintain transparency and accountability.

4.4. Geographical Diversification

4.4.1. Finland (36% of Portfolio)

Finland remains the core focus of the Borealis Wealth Fund. Domestic exposure is concentrated in green infrastructure, forestry, and conservation assets that directly contribute to national restoration goals. These investments accelerate Finland's transition toward a circular bioeconomy, strengthen its carbon sink potential, and improve resilience in key ecosystems such as wetlands and boreal forests.

4.4.2. European Union (10% of Portfolio)

The fund allocates a moderate share of capital to other European Union markets to align with EU-level sustainability standards and leverage initiatives under the European Green Deal and the Nature Restoration Law. Exposure is primarily through private credit, public equities, and conservation projects that benefit from transparent regulatory frameworks and well-developed environmental markets.

4.4.3. Global Exposure (54% of Portfolio)

The global component of the portfolio provides diversification and growth opportunities across different bioregions. Forestry holdings in Argentina and Laos expand the fund's access to scalable reforestation projects and high-yield timberlands, complementing Finland's mature forest base. Global private equity, green bonds, and biodiversity-linked credit instruments broaden the fund's reach to innovative sustainability ventures and nature-based markets in both developed and emerging economies.

4.4.4. Cash (5% of Portfolio)

Liquidity reserves are fully held in Finland to ensure accessibility for short-term obligations and operational flexibility.

Geographical Diversification

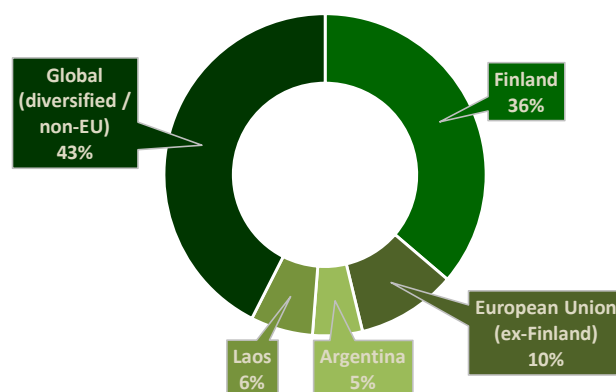


Figure 7.

5. Performance Measurement and Benchmarking

5.1. Overview of the Benchmarking Framework

The Borealis Wealth Fund (BWF) operates under a dual mandate that prioritizes both long-term financial performance and measurable biodiversity restoration. To ensure accountability across these dimensions, the fund adopts a two-tier benchmarking framework. The first-tier measures financial returns relative to global and sectoral indices, while the second tier evaluates biodiversity and nature-related outcomes. Together, these benchmarks provide a structured framework for aligning financial stewardship with ecological impact.

Traditional investment benchmarks, while effective for return attribution, fail to capture the externalities or positive spillovers of nature-positive capital allocation. The BWF's framework addresses this gap by linking asset class performance to measurable sustainability criteria, including biodiversity outcomes, sustainable management practices, governance quality, and exposure to nature-related risks. This approach allows the fund to report transparently on both return and impact metrics, in line with emerging standards under the Taskforce on Nature-related Financial Disclosures (TNFD) and the EU Sustainable Finance Disclosure Regulation (SFDR).

The fund's performance is reviewed annually, with both financial and biodiversity benchmarks recalibrated to reflect market developments and policy shifts. This ensures that capital allocation remains adaptive to evolving global sustainability objectives while maintaining disciplined financial performance.

5.2. Financial Benchmark Construction

The financial benchmark for the Borealis Wealth Fund is designed as a weighted composite index that mirrors the fund's long-term strategic asset allocation. Each component index was selected to provide a representative measure of risk, return, and sustainability alignment within its respective asset class. The benchmark is intended to evaluate the portfolio's performance over a 10-year horizon, providing a consistent reference point for assessing strategic progress toward both profitability and resilience. (refer to Table 1)

5.3. Biodiversity Impact Benchmark

The Biodiversity Impact Benchmark complements the financial benchmark by assessing how effectively the fund's investments deliver measurable ecological outcomes. It translates the Borealis Wealth Fund's dual mandate into a quantifiable and transparent framework for screening and monitoring all nature-linked investments.

Table 1: Borealis Wealth Fund's Financial Benchmark

Borealis Wealth Fund's Benchmark	
Benchmarks	Weights
NCREIF Timberland Property Index	25%
Dow Jones Brookfield Global Green Infrastructure Index	20%
Bloomberg Barclays MSCI Global Green Bond Index	15%
iShares Global Agriculture Index ETF	15%
MSCI World Index	10%
S&P Global Carbon Credit Indices Methodology	10%
S&P U.S. Treasury Bill Index	5%
Total	100%

The Biodiversity Impact Score (BIS) serves as the primary evaluation tool, integrating the five previously defined indicators: Biodiversity Outcomes, Climate Co-benefits, Sustainable Management Practices, Governance & Disclosure, and Nature-related Risk. Each indicator captures a distinct dimension of ecological performance and is weighted according to its relevance to long-term biodiversity restoration. (Refer to Figure 2 or Appendix 8.1)

6. Impacts and Expected Outcomes

6.1. Financial Impact

Over the forecast horizon to 2035, the fund’s AUM is projected to grow from €10B in 2024 to approximately €38B by 2035, assuming a blended return of 6-7% annually under the base scenario.

Real Assets, particularly forestry, land holdings, and green infrastructure, serve as the primary engine of value creation, contributing roughly half of the AUM growth. The introduction of private credit and equity is expected to enhance returns while maintaining portfolio resilience. Strong tailwinds underpin these returns from:

- Rising global demand for sustainable timber, engineered wood, and green materials
- Expansion in renewable energy infrastructure across Europe
- Institutional interest in biodiversity-aligned investments, providing potential co-financing and liquidity opportunities

6.2. Environmental Impact

The BWF’s investment strategy directly supports Finland’s national environmental goals and the European Union’s green policies. In particular, the fund aligns its ecological targets with the EU Biodiversity Strategy for 2030, the EU Nature Restoration Law, and the EU Taxonomy for Sustainable Activities, while contributing to Finland’s 2035 Carbon Neutrality Act. These are the Key Projected Outcomes by 2035:

- Restoration or protection of over 500,000 hectares of forests, peatlands, and wetlands across Finland and partner countries, contributing directly to the EU’s target of restoring at least 20% of degraded ecosystems by 2030.
- Net sequestration exceeding 6-8 million tonnes of CO₂ annually, positioning Finland to reestablish its role as a net carbon sink and advance toward its carbon-neutral 2035 goal under national legislation.
- Reduction in biodiversity risk exposure by 25-30% , achieved through certified forest management, nature-based infrastructure investments, and compliance with TNFD-aligned disclosure frameworks.

7. Risks and Mitigation Strategies

7.1. Key Risks

- **Market and Macroeconomic Risks**
Volatility in global commodity markets, interest rates, and exchange rates can affect the valuation of forestry, infrastructure and private credit assets
- **Climate and Physical Environmental Risks**
Climate change introduces physical risks to forestry and land holdings, including increased wildfire frequency, pest outbreaks, and shifts in species composition. These events may reduce forest productivity or disrupt carbon sequestration targets.
- **Regulatory and Policy Risks**
The EU’s evolving sustainability framework, covering SFDR, CSRD, and the Nature Restoration Law, imposes complex disclosure and compliance requirements. Misalignment or delayed implementation could result in restricted market access or reputational damage.
- **Biodiversity and Data Integrity Risks**
Accurate measurement of biodiversity outcomes remains challenging due to data gaps, inconsistent baselines, and limited comparability across ecosystems. Poor data quality could undermine the credibility of impact reporting and the Biodiversity Impact Score (BIS).
- **Transition and Technological Risks**
As the global economy decarbonizes, shifts in technology and consumer preferences may alter the profitability of certain investments. For instance, bio-based materials may face competition from new synthetic substitutes, while green infrastructure projects may be exposed to high upfront capital costs.
- **Governance and Operational Risks**
The complexity of multi-sector investments introduces risks in execution, stakeholder engagement, and cross-border project management. Inadequate oversight or partner misalignment could jeopardize both ecological and financial outcomes.

7.2. Risk Mitigation

Risk Type	Mitigation Strategy
Market & Macroeconomic	Diversify across asset classes, geographies, and counterparties; maintain a 5 percent liquidity buffer; conduct annual stress testing using macroeconomic scenarios from the Bank of Finland and IMF.
Climate & Physical Environmental	Apply TNFD-aligned risk mapping (LEAP framework) for all nature-dependent assets; invest in adaptive silviculture and fire prevention; secure parametric insurance for forestry losses.
Regulatory & Policy	Establish a regulatory compliance team; ensure full Taxonomy and SFDR alignment; maintain real-time policy tracking with EU and Finnish Environment Ministry updates; adopt double materiality reporting consistent with CSRD.
Biodiversity & Data Integrity	Partner with LUKE, WWF Finland, and the Finnish Environment Institute for third-party data validation; deploy remote sensing and satellite monitoring to verify restoration outcomes annually.
Transition & Technological	Prioritize scalable, low-carbon innovations such as engineered wood (LVL, CLT) and bioenergy; maintain a flexible investment horizon to reallocate capital based on technological shifts.
Governance & Operational	Implement multi-tier governance: Investment Committee for financial oversight and Biodiversity Oversight Committee for ecological performance; conduct annual dual mandate audits with independent assurance providers.

8. Appendix

8.1. Biodiversity Measures

1) Biodiversity Outcomes: This variable helps us quantify the percentage of the asset's land is protected or actively restored. The formula for Biodiversity can be the following:

$$\text{Generic Biodiversity Impact (in \%)} = \frac{\text{Biodiversity Positive Area or Activity}}{\text{Total Relevant Area or Activity}} * 100$$

$$\text{Forestry Specific Biodiversity Impact (in \%)} = \frac{\text{Hectares Protected or Restored}}{\text{Total Managed Hectares}} * 100$$

Sector examples:

Forestry: Hectares of forest under protection/restoration ÷ total managed hectares.

Agri-Tech: Farmland under regenerative practices (e.g., cover crops, hedgerows, pollinator strips) ÷ total farmland.

Materials: % of raw wood sourced from suppliers that set aside conservation areas.

Sovereigns: % of national land under formal protection/restoration.

Corporates: % of supply chain volume sourced from biodiversity-positive regions or initiatives.

Benchmarking:

<20% = Weak, 20-50% = Moderate, >50% = Strong alignment with biodiversity goals.

2) Climate Co-benefits: Net Carbon Impact of the asset, carbon sinks also seem to be a problem, which needs to be addressed by the fund.

Net Carbon Balance = Carbon Sequestration - Carbon Emissions

Sector examples:

Forestry: CO₂ sequestered in standing forests – logging/transport emissions.

Agri-Tech: Soil carbon gains – emissions from fertilizers, fuel, and livestock.

LVL/Materials: Lifecycle carbon intensity of product (kg CO₂e/m³ wood vs. steel/concrete alternatives).

Sovereigns: National GHG emissions/removals (forestry & land-use sector balance).

Corporates: Portfolio Weighted Average Carbon Intensity (WACI) with ecosystem offsets considered.

Benchmarking:

Negative (carbon sink) = Leader, Neutral (±7% of 0) = Aligned, Positive (Net Source) = At Risk

3) Sustainable Management Practices: How much of the managed area meets credible forestry standards, the certifications ensure biodiversity, soil and water standards in daily management.

$$\text{Generic Certified Area (in \%)} = \left(\frac{\text{Certified or Verified Units}}{\text{Total Units}} \right) * 100$$

$$\text{Forestry Certified Area (in \%)} = \left(\frac{\text{FSC or PEFC Certified Hectares}}{\text{Total Hectares Managed}} \right) * 100$$

Sector examples:

Forestry: FSC/PEFC certified hectares ÷ total hectares managed.

Agri-Tech: % of farmland certified Organic, Rainforest Alliance, or regenerative agriculture.

LVL/Materials: % of timber inputs FSC/PEFC certified, or suppliers audited for ESG.

Sovereigns: Ratification and implementation of biodiversity treaties (CBD, GBF, CITES).

Corporates: % of suppliers covered by verified sustainability certifications.

Benchmarking:

>85% = Leader, 50% < n < 85% = Aligned, <50% = At Risk

4) Governance & Disclosure: This enables accountability and comparability across assets. It also measures the transparency, policies and biodiversity risk reporting.

$$\text{Disclosure Score} = \sum (\text{Policy Transparency} + \text{Alignment with TNFD or SFDR reporting frameworks} + \text{Independent assurance of biodiversity disclosures} + \text{Stakeholder engagement})$$

Scoring rubric (use this for each of the 4 items):

0 pts (Absent): No policy/reporting/assurance/engagement, or only vague statements.

12.5 pts (Partial): Some elements exist, but **not** comprehensive, not fully public, or only partially cover biodiversity; ad-hoc engagement; internal review only.

25 pts (Full): Clear, public, biodiversity-specific policy/reporting; TNFD or SFDR mapped, independent assurance, structured engagement with evidence and outcomes.

Benchmarking:

>70 = Leader, 40-70 = Aligned, <40 = At Risk

5) Nature Related Risks: This measures the dependence of an asset on an ecosystem.

$$\text{Nature Related Risk} = 100 - \left(\frac{\text{Exposure to nature related risks}}{\text{Sector or Geography Benchmark}} * 100 \right)$$

Sector examples:

Forestry: % of land at high fire/pest/erosion risk.

Agri-tech: % production dependent on pollinators, water stress.

LVL/Materials: % raw inputs from high-deforestation regions.

Sovereigns: % GDP in nature-dependent sectors.

Corporates: % revenue in high-risk industries (from ENCORE/TNFD mapping).

What the numbers mean:

If asset risk < benchmark - Positive (>50), This means low exposure vs peers

If asset risk ≈ benchmark - Neutral (around 50) average exposure

If asset risk > benchmark - Lower Score (<50) high exposure vs peers

Benchmarking:

>70 = Leader, 40-69 = Aligned, <40 = At Risk

Indicator	What it Measures (Simple Explanation)	Weighting
Biodiversity Outcomes	% of land or activity that is protected, restored, or biodiversity-positive. Captures the direct ecological contribution.	30.00%
Climate Co-benefits	Whether the asset is a net carbon sink or source. Balances carbon sequestration vs. emissions.	20.00%
Sustainable Management Practices	% of operations, land, or inputs covered by credible sustainability certifications (FSC, PEFC, Organic, etc.). Ensures daily practices meet biodiversity and soil/water standards.	15.00%
Governance & Disclosure	Strength of policies, transparency, reporting, assurance, and stakeholder engagement related to biodiversity. Ensures accountability and comparability.	15.00%
Nature Related Risk	Degree to which the asset depends on ecosystem services (water, soil, pollinators, forests) and is exposed to nature-related risks.	20.00%

8.2. Lumber Growth Projections

Lumber Growth			
Types of Lumber	Conservative CAGR (till 2035)	Base CAGR (till 2035)	Best CAGR (till 2035)
Softwood Lumber	1.00%	2.00%	3.50%
Sawn Timber (all grades)	1.00%	2.00%	3.00%
Global Lumber	0.90%	2.00%	3.50%
Timber/Engineered Wood/Construction	6.00%	7.00%	9.00%
Wood Based Panels/ Engineered Wood	2.00%	3.00%	4.00%
Mean	1.61%	2.79%	4.21%

8.3. AUM Growth Projections With Constant 6% CAGR

AUM With 6% Return	2025F	2026F	2027F	2028F	2029F	2030F	2031F	2032F	2033F	2034F	2035F
Conservative AUM (in B)	11.60 €	13.30 €	15.09 €	17.00 €	19.02 €	21.16 €	23.43 €	25.84 €	28.39 €	31.10 €	33.97 €
Base AUM (in B)	11.60 €	13.30 €	15.09 €	17.00 €	19.02 €	21.17 €	23.44 €	25.84 €	28.40 €	31.10 €	33.97 €
Best AUM (in B)	11.60 €	13.30 €	15.10 €	17.00 €	19.02 €	21.17 €	23.44 €	25.85 €	28.40 €	31.11 €	33.98 €

8.4. AUM Growth Projections With Constant 6% CAGR (2025-2031) then 7.47% (2032-2035)

AUM With Expected Returns	2025F	2026F	2027F	2028F	2029F	2030F	2031F	2032F	2033F	2034F	2035F
Conservative AUM (in B)	11.60 €	13.30 €	15.09 €	17.00 €	19.02 €	21.16 €	23.43 €	26.18 €	29.14 €	32.32 €	35.73 €
Base AUM (in B)	11.60 €	13.30 €	15.09 €	17.00 €	19.02 €	21.17 €	23.44 €	26.19 €	29.15 €	32.32 €	35.74 €
Best AUM (in B)	11.60 €	13.30 €	15.10 €	17.00 €	19.02 €	21.17 €	23.44 €	26.19 €	29.15 €	32.33 €	35.75 €

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